Modelling Book Auctions: Catalogues & Large Language Models

Fox, Marika Kyranna
marika.fox@uantwerpen.be
University of Antwerp, Belgium

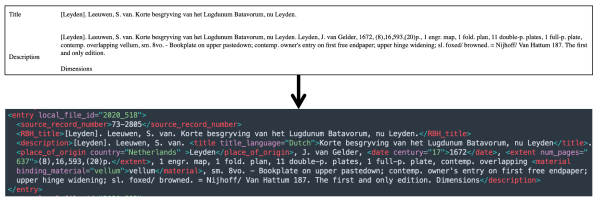
Auction prices act as a useful proxy for the cultural value of objects that have both intrinsic and socially perceived value, because they represent how much a buyer is willing to pay at a specific point in time. And since auction houses typically estimate the worth of items pre-sale, the final price can be compared with expert predictions that are based on available information about the item but cannot consider external factors such as personal connections to buyers. This sometimes leads to items failing to meet their minimum price and going unsold- or fetching a price far above the high estimate. See Figure 1 for an example of this occurring in the case of a minuscule prayer book. This project seeks to use computational analysis of large sets of auction price data to eventually build a model that can more accurately predict such prices.

The catalogue entry (above) for Lot 943 of the Arenberg Auctions December 2021 sale, showing an estimated price of 1000-1500€, compared to the recorded hammer price (below) of 3500€ ("Arenberg Auctions Auction Results" n.d.) .

Previous work on modelling auction price trends has focused largely on the art market (Dimson / Spaenjers 2014; Goetzmann 1993) . However, very little prior work has been done on book sales; there are some case studies on specific dealers, buyers, and items (Botana / Cleaver 2024) , and explanations of shorter-term trends in the book market (Lehmann-Haupt 1961) , but no existing large-scale datasets or experimental results.

The first stage of this project, currently in progress, is focused on the challenge of extracting and structuring useful data from free-text descriptions of manuscripts and early books that have appeared in auction catalogues. Large language models such as OpenAI's ChatGPT have proven helpful as data annotators in similar tasks (Karjus 2023) , and that is the method that is currently explored in this project. 100 lot descriptions copied from the RareBookHub.com database ("Rare Book Hub" n.d.) are being used for proof of concept.

Two methods of annotation have been explored so far. The first is the addition to each description of XML tags that identify information about the item's size, material, place of origin, language, date, and title. The data itself was recorded as attributes within each tag. This task was performed by me, one other expert human annotator, and finally OpenAI's GPT4, according to guidelines written by me. See Figure 2 for an example of an annotated description.



An example of a single lot description from the Rare Book Hub (above) and the same description in XML format after the relevant information has been tagged (below)

Table 1 shows two measures of inter-annotator agreement between the two human annotators and GPT4 for this task, considering the placement of the tags at a character level. The consistently high values show that overall, the use of my written guidelines and GPT4 to curate a usable dataset is a viable method.

Table 1 : Character-level inter-annotator agreement for XML tagging of 100 sample auction catalogue entries

Annotator Pair	Overlap	Cohen's Kappa
Annotator 1 & Annotator 2	0.952	0.797
Annotator 1 & GPT4	0.929	0.726
Annotator 2 & GPT4	0.924	0.715

The process of adding tags to the description text indicates that GPT4 is not only able to extract the data that I need, but is also able to identify exactly where its data is coming from. However, this method does have certain downsides. The first is that the model's output was inconsistent: sometimes it would alter the original description text, and sometimes it would tag categories other than the requested ones (see Figure 3). The other is that the

need to place the tags limited the amount of data that could be recorded. For example, many fragments are heavily implied to consist of a single leaf, but there is no explicit statement to that end that could be tagged.

0	84871
title	6476
extent	1907
date	1103
material	949
place_of_origin	793
dimensions	742
language	348
Name: tag, dtype: int64	

0	81628
title	7827
extent	2950
material	1338
date	1122
dimensions	1115
place_of_origin	824
language	319
editor	34
author	32
Name: tag, dtype: int64	

Counts of characters contained within each category of tag in my annotations (left) and GPT4's annotations (right). Note that GPT4 included categories for editors and authors, despite the prompt requesting that it use only the same categories as I did

To solve these issues, the next method tested was requesting that GPT4, for each description, return only the data in the form of a JSON schema, which restricted it into only considering the desired categories and allowed it to infer some information such as the extent of single-leaf fragments. I then formatted my collected data as a csv table for comparison. Unfortunately, the overlap between my results and the model's' with this method was at first worse than the overlap between mine and the model's data as recorded earlier in the form of XML attributes (see Figure 5). Closer inspection showed that the model was erring on the side of making too many inferences, such as recording that certain books were written on paper rather than parchment even if there was no mention of either in the catalogue description. Prompt engineering thus far has focused on changes to the guidelines for each individual data category, with the hope of better aligning mine and GPT4's underlying assumptions about what to record (see Figure 4 for an example of this).

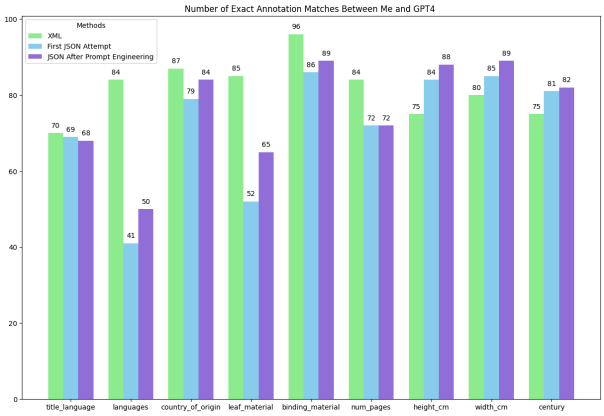
```
leaf_material: Material of the leaves, e.g. parchment or paper. Record as given (minus any spelling mistakes): retain
distinction between, for instance, vellum and parchment. If both are used in the description, note both
```

↓

```
leaf_material: Material of the leaves, e.g. parchment or paper. Record as given (minus any spelling mistakes): retain
distinction between, for instance, vellum and parchment. If both are used in the description, note both. If the leaf material is
not explicitly mentioned, do not record anything
```

An extract from the annotation guidelines for the leaf_material category, before (above) and after (below) prompt engineering. The addition of the last sentence helped to discourage GPT4 from defaulting to recording "paper" if the description did not specify

Figure 5 provides an overview of the comparison between my and GPT4's annotations for the XML format, the initial run of the JSON format, and the most recent run of the JSON format after some prompt engineering. There is still clearly room for improvement, but GPT4 is proving to be an effective annotation assistant that will be invaluable as I start working with much larger amounts of data.



The number of entries out of the collection of 100 samples for which I and GPT4 recorded the exact same information for the given category

Bibliography

"Arenberg Auctions Auction Results" (n.d.): Arenberg Auctions <https://www.my.arenbergauctions.com/en/archive> [29.01.2024].

Botana, Federico / Cleaver, Laura, eds. (2024): *The Economics of the Manuscript and Rare Book Trade, ca. 1890–1939* . Leeds: Arc Humanities Press.

Dimson, Elroy / Spaenjers, Cristophe (2014): "Investing in Emotional Assets". SSRN Scholarly Paper. Rochester, NY.

Goetzmann, William N. (1993): "Accounting for Taste: Art and the Financial Markets Over Three Centuries", in: *The American Economic Review* 83, 5: 1370–76.

Karjus, Andres (2023): "Machine-Assisted Mixed Methods: Augmenting Humanities and Social Sciences with Artificial Intelligence". arXiv <http://arxiv.org/abs/2309.14379> [29.04.2025].

Lehmann-Haupt, Hellmut (1961): "Current Trends in Antiquarian Books", in: *Library Trends* 9, 4 <https://hdl.handle.net/2142/5971> [29.04.2025].

"Rare Book Hub" (n.d.): <https://www.rarebookhub.com/> [29.04.2025].